

Supplementary materials

Table S1 The number of neurons recorded in each session for WT and 5xFAD groups.

5xFAD				
Mouse index	session 1	session 2	session 3	session 4
1	411	336	—	408
16	295	276	—	288
117	36	73	78	94
118	27	17	28	—
123	80	68	128	—
WT				
3	378	381	295	564
5	413	—	574	352
7	—	196	627	420
102	173	72	—	47
106	106	341	82	—
109	144	97	125	108
110	284	244	—	275

Cross-validation

During fine-tuning, cross-validation was performed to test the hypothesis regarding the influence of individual session quality on the final results.

Cross-entropy loss of each fold is shown in [Figure S1](#). The F1 score is shown in [Figure S2](#). The plots are presented in a table, where each cell contains the curve for a specific fold; columns are organized by the transgenic subject ID, and rows by the wild-type subject. These plots demonstrate that the network trains stably, with the exception of some folds involving mouse 106. The average accuracy and standard deviation for each session of the validation sample calculated for the entire cross-validation experiment is shown in the [Table S2](#).

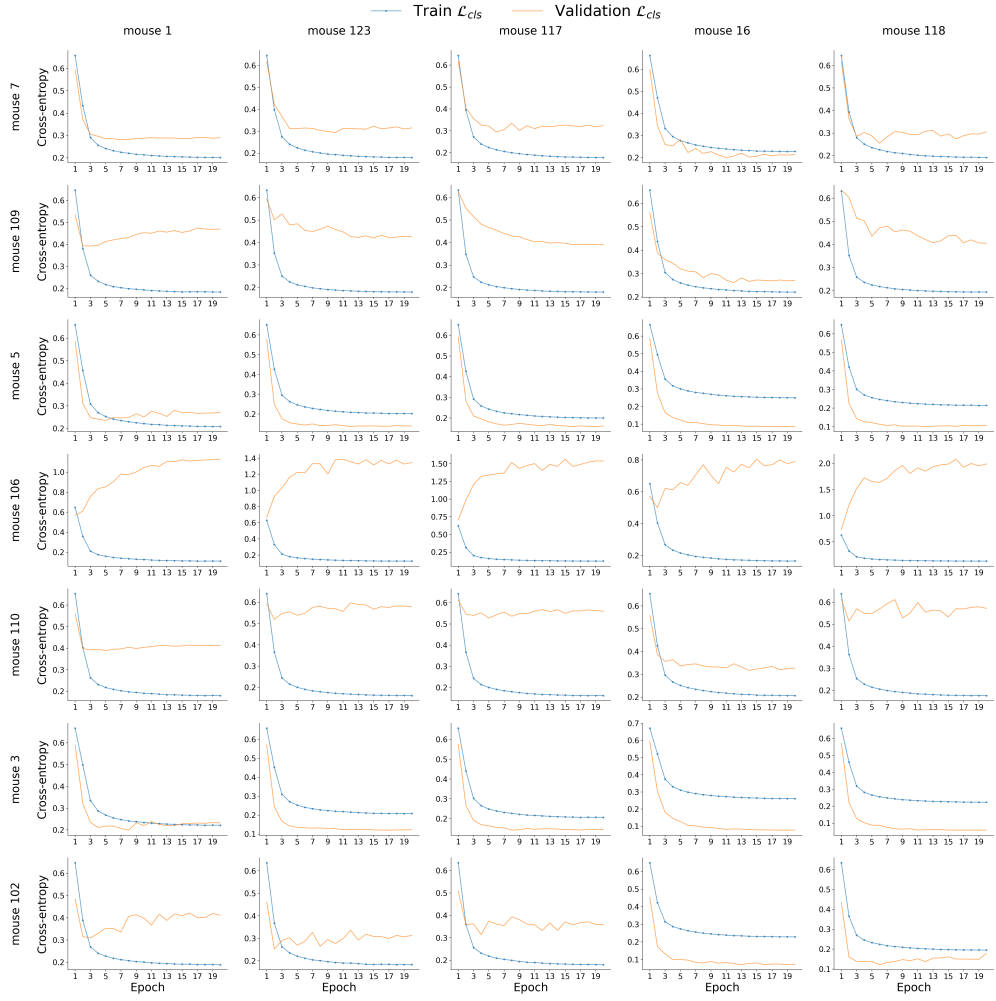


Fig. S1 Cross-entropy loss for each fold in the cross-validation experiment. Each graph shows the loss functions by epochs for the training and validation subsample. Each row is signed on the left and corresponds to one wild-type individual. Each column is signed at the top and corresponds to a transgenic individual. The column signature and the lines of each graph show which two individuals form the same validation sample.

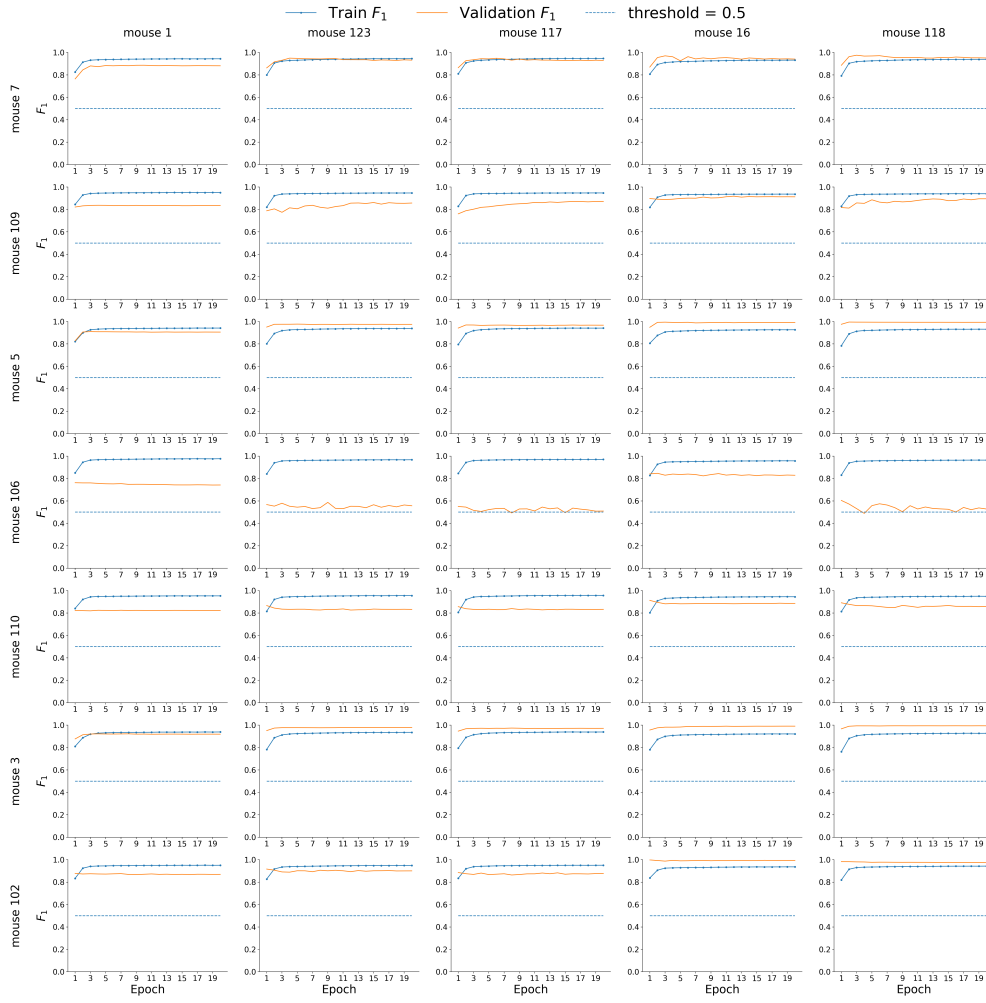


Fig. S2 Curves of the F1 score for each fold in the cross-validation experiment. Each graph shows the F1 score by epochs for the training and validation subsample.

Table S2 Model prediction accuracy on individual sessions for each mouse during cross-validation. Results are reported as the mean $\pm$ standard deviation across all cases in which the mouse was included in the validation set.

5xFAD				
Mouse index	session 1	session 2	session 3	session 4
1	0.9991 $\pm$ 0.0007	0.2219 $\pm$ 0.0220	—	0.9993 $\pm$ 0.0005
16	0.9685 $\pm$ 0.0133	0.9864 $\pm$ 0.0059	—	1.0000 $\pm$ 0.0000
117	0.2682 $\pm$ 0.0869	0.3269 $\pm$ 0.0507	0.9905 $\pm$ 0.0046	0.9174 $\pm$ 0.0426
118	0.6092 $\pm$ 0.0674	0.3352 $\pm$ 0.1178	0.9982 $\pm$ 0.0025	—
123	0.7880 $\pm$ 0.0299	0.2997 $\pm$ 0.0789	0.9993 $\pm$ 0.0007	—
WT				
3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9901 $\pm$ 0.0096	0.9973 $\pm$ 0.0008
5	0.9980 $\pm$ 0.0020	—	0.9971 $\pm$ 0.0021	1.0000 $\pm$ 0.0000
7	—	0.6449 $\pm$ 0.0493	0.9799 $\pm$ 0.0034	0.9433 $\pm$ 0.0136
102	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	—	1.0000 $\pm$ 0.0000
106	0.8282 $\pm$ 0.0625	0.2150 $\pm$ 0.0320	0.4926 $\pm$ 0.0448	—
109	0.8042 $\pm$ 0.0795	0.8580 $\pm$ 0.0597	0.8864 $\pm$ 0.0708	0.6930 $\pm$ 0.0811
110	0.9986 $\pm$ 0.0015	0.9989 $\pm$ 0.0017	—	0.3175 $\pm$ 0.0328

In this experiment, 0.5 was chosen as the threshold value for the following reasons. Each activity matrix actually represents the activity of one neuron, provided with some additional statistics about its environment. Thus, the proportion of matrices obtained from one session to a specific class can act as an indirect metric of the model’s confidence in predicting the entire session. If more than half of the data from one session is assigned by the model to a specific class, then it is classified to this class.

For all folds used in cross-validation, we plotted the evolution of prediction accuracy for each individual mouse session in the validation set. The resulting graph is shown in [Figure S3](#).

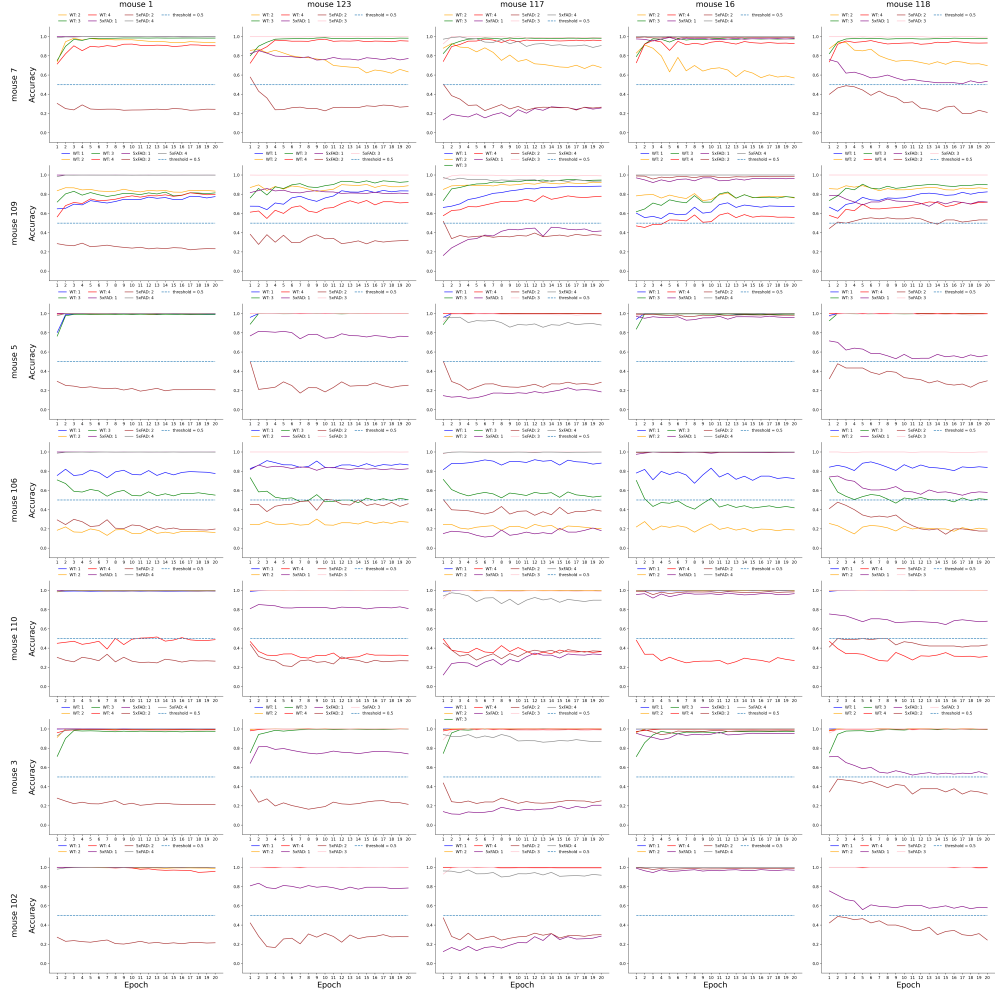


Fig. S3 Classification accuracy across validation sessions for each fold. Each graph shows the dependence of classification accuracy on the epoch number. The threshold is equal to 0.5.

In the cross-validation experiment, a multitask learning approach was employed. [Table S3](#) presents the final metrics obtained from the cross-validation experiment for the validation set.

Table S3 Cross-validation performance metrics for the validation set using a multitask learning approach. The table shows the results for each combination of mouse indices from the wild-type (WT) and 5xFAD groups.

WT	5xFAD	R ²	MAE	cosine similarity	precision	recall
7	1	0.9868 ± 0.0049	0.0051 ± 0.0024	0.9922 ± 0.0140	0.9499	0.8224
7	123	0.9794 ± 0.0056	0.0052 ± 0.0024	0.9927 ± 0.0121	0.9182	0.9464
7	117	0.9783 ± 0.0054	0.0051 ± 0.0023	0.9922 ± 0.0131	0.9269	0.9316
7	16	0.9852 ± 0.0053	0.0052 ± 0.0026	0.9930 ± 0.0123	0.8949	0.9929
7	118	0.9786 ± 0.0053	0.0050 ± 0.0023	0.9923 ± 0.0129	0.9190	0.9858
109	1	0.9821 ± 0.0047	0.0061 ± 0.0025	0.9921 ± 0.0125	0.8028	0.5944
109	123	0.9647 ± 0.0085	0.0076 ± 0.0024	0.9927 ± 0.0080	0.8437	0.8713
109	117	0.9613 ± 0.0102	0.0075 ± 0.0023	0.9919 ± 0.0096	0.8849	0.8578
109	16	0.9782 ± 0.0049	0.0064 ± 0.0027	0.9932 ± 0.0098	0.6900	0.9579
109	118	0.9553 ± 0.0128	0.0081 ± 0.0023	0.9920 ± 0.0076	0.8272	0.9755
5	1	0.9783 ± 0.0080	0.0060 ± 0.0026	0.9923 ± 0.0126	0.9943	0.8322
5	123	0.9576 ± 0.0132	0.0066 ± 0.0026	0.9927 ± 0.0103	0.9991	0.9525
5	117	0.9533 ± 0.0143	0.0065 ± 0.0025	0.9923 ± 0.0111	0.9993	0.9380
5	16	0.9739 ± 0.0095	0.0061 ± 0.0027	0.9931 ± 0.0108	0.9957	0.9878
5	118	0.9474 ± 0.0158	0.0066 ± 0.0026	0.9924 ± 0.0107	0.9986	0.9888
106	1	0.9851 ± 0.0048	0.0058 ± 0.0026	0.9914 ± 0.0152	0.3412	0.3965
106	123	0.9751 ± 0.0073	0.0062 ± 0.0027	0.9923 ± 0.0119	0.4217	0.8201
106	117	0.9730 ± 0.0080	0.0061 ± 0.0026	0.9913 ± 0.0140	0.3881	0.7422
106	16	0.9825 ± 0.0054	0.0058 ± 0.0029	0.9927 ± 0.0125	0.3283	0.9901
106	118	0.9735 ± 0.0079	0.0061 ± 0.0027	0.9914 ± 0.0134	0.3696	0.9240
110	1	0.9838 ± 0.0057	0.0057 ± 0.0025	0.9924 ± 0.0129	0.8202	0.7261
110	123	0.9714 ± 0.0095	0.0064 ± 0.0024	0.9933 ± 0.0096	0.7677	0.9086
110	117	0.9687 ± 0.0102	0.0063 ± 0.0023	0.9928 ± 0.0106	0.7815	0.8917
110	16	0.9808 ± 0.0068	0.0059 ± 0.0027	0.9936 ± 0.0103	0.7473	0.9771
110	118	0.9670 ± 0.0110	0.0064 ± 0.0023	0.9932 ± 0.0095	0.7640	0.9815
3	1	0.9853 ± 0.0059	0.0054 ± 0.0022	0.9892 ± 0.0187	0.9913	0.8574
3	123	0.9764 ± 0.0075	0.0057 ± 0.0021	0.9885 ± 0.0186	0.9988	0.9579
3	117	0.9748 ± 0.0075	0.0056 ± 0.0020	0.9881 ± 0.0195	0.9984	0.9467
3	16	0.9833 ± 0.0063	0.0055 ± 0.0023	0.9896 ± 0.0175	0.9939	0.9881
3	118	0.9737 ± 0.0074	0.0057 ± 0.0020	0.9871 ± 0.0210	0.9979	0.9905
102	1	0.9880 ± 0.0047	0.0055 ± 0.0023	0.9882 ± 0.0203	0.9905	0.5167
102	123	0.9779 ± 0.0100	0.0065 ± 0.0021	0.9829 ± 0.0263	1.0000	0.8185
102	117	0.9745 ± 0.0116	0.0062 ± 0.0019	0.9819 ± 0.0283	1.0000	0.7800
102	16	0.9857 ± 0.0060	0.0057 ± 0.0026	0.9892 ± 0.0177	1.0000	0.9596
102	118	0.9678 ± 0.0155	0.0065 ± 0.0019	0.9757 ± 0.0373	1.0000	0.9491

Baseline Experiment

To evaluate the performance of the presented model, we conducted a comparison against baseline: a recurrent neural network (RNN), a logistic regression classifier. Also, the final version of NEuRT performance was compared with itself without pre-train stage. The RNN baseline shared the same output head and dimensionality as our main model. The logistic regression baseline was implemented by flattening the input channels (512×5) into a single vector of size 2560, followed by a linear layer without activation.

Table S4 Models’ performance comparison on the reconstruction and classification tasks. The same test sample and seed was used for all models test. The last training epoch results were used.

	Reconstruction				
Channel	Signal	μ_{close}	σ_{close}	μ_{far}	σ_{far}
	R^2				
NEuRT with pretrain	0.9766	0.9827	0.9595	0.9836	0.9662
NEuRT without pretrain	0.4918	0.6398	0.3901	0.7284	0.4120
RNN	0.7535	0.0015	-0.0650	-0.0434	-0.0729
LogisticRegression	NaN	NaN	NaN	NaN	NaN
	MAE				
NEuRT with pretrain	0.0112	0.0047	0.0065	0.0036	0.0046
NEuRT without pretrain	0.0893	0.0277	0.0334	0.0185	0.0257
RNN	0.0607	0.0471	0.0433	0.0365	0.0330
LogisticRegression	NaN	NaN	NaN	NaN	NaN
	cosine similarity				
NEuRT with pretrain	0.9714	0.9982	0.9976	0.9990	0.9990
NEuRT without pretrain	0.6589	0.9646	0.9664	0.9865	0.9848
RNN	0.8089	0.9675	0.9689	0.9881	0.9863
LogisticRegression	NaN	NaN	NaN	NaN	NaN
	Classification				
	precision		recall		
NEuRT with pretrain	0.9948		0.9850		
NEuRT without pretrain	0.7875		0.8790		
RNN	0.6450		0.6488		
LogisticRegression	0.9963		0.9507		

The baseline experiment highlights the comparative performance of the RNN, logistic regression, and NEuRT models across reconstruction and classification tasks. [Table S4](#) demonstrates that the NEuRT model with pretraining consistently outperforms the other baselines, achieving the highest scores in R^2 , MAE, and cosine similarity.

Interpretation methods

In Figure S4, the confusion matrices obtained by zeroing out individual components of the activity matrix are shown.

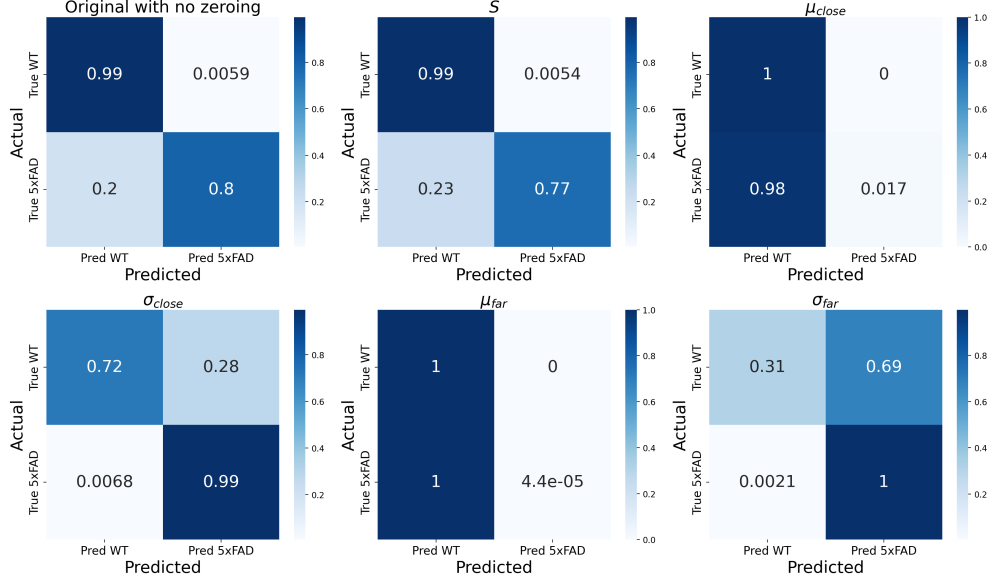


Fig. S4 Confusion matrices on the slice level for the model outputs with zeroing activity matrix's individual components. The color gradient indicates results normalized to the true value.

To estimate the importance of temporal segments, we employed an adapted attention rollout method, while also testing simplified and extended variants. Two modifications were considered. The first involved using entropy within a single attention layer to reduce the dominance of diagonal attention matrices:

$$W_l = \text{softmax}\left(\frac{1}{L} \sum_{i=1}^L \left(-\sum_{j=1}^L A_{tij}^h \log_2 (A_{tij}^h + 10^{-12})\right)\right)$$

$$\overline{A}_l^h = W_l^h \cdot A_l^h$$

Instead of the standard averaging over attention heads within a layer, a weighted average was applied.

The second modification involved using gradients as weights, not for entire matrices, but for individual elements. A similar approach was used in (Chefer et al., 2021), where relevance weights were applied instead of attention matrices. The gradient-based weighting is expressed as:

$$\overline{A}_l^h = \nabla A_l^h \odot A_l^h$$

We compared variants using entropy alone and entropy combined with gradients. The most notable behavior, however, was observed when using only the pooling vector weights to estimate temporal importance, as shown in [Figure S5](#), where methods are compared in the top-k zeroing experiments described in the Results.

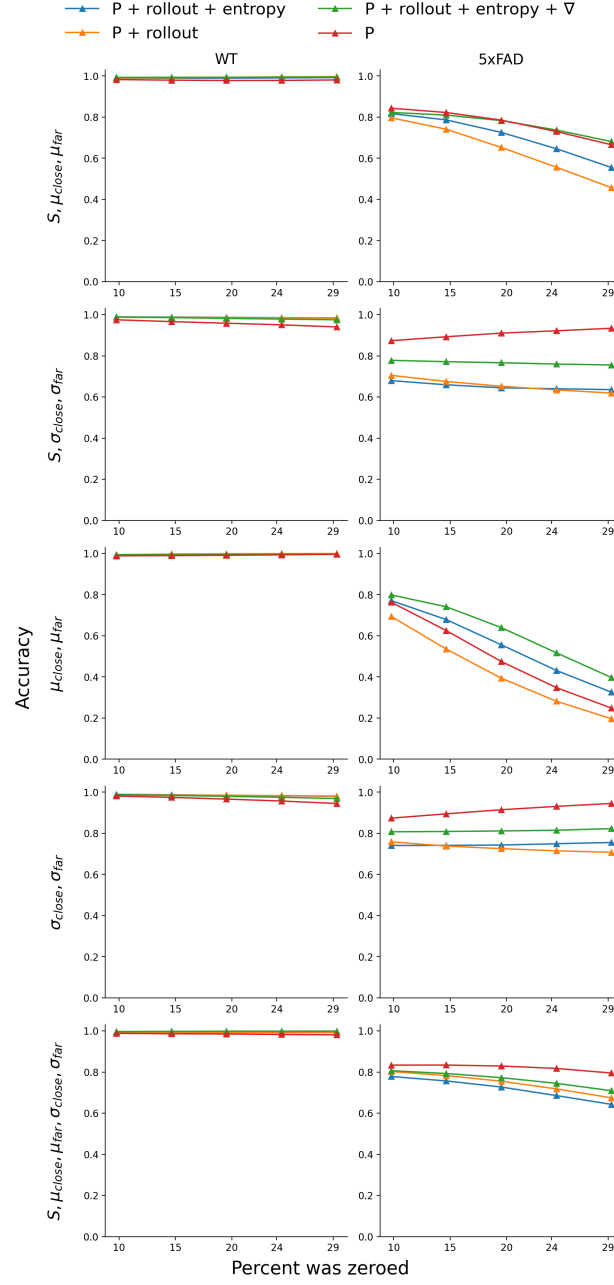


Fig. S5 Comparison of the model's prediction accuracy for both classes when zeroing out the top-k temporal segments of activity matrix component groups based on their importance using different interpretation methods: P + rollout + entropy – a combination of pooling weights and entropy-weighted attention rollout; P + rollout – a combination of pooling weights and unweighted attention rollout; P + rollout + entropy + ∇ – a combination of pooling weights, entropy-weighted attention rollout, and gradients; P – pooling weights alone.